

Devin Ha

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Work Experience

ML Engineer

aignostics

04/2026 – Present

Berlin, Germany

- Develop models for Digital Pathology using large vision foundation models.
- Training of models for cell detection and tissue segmentation
- Benchmarking of those models

Technical Lead and ML Engineer

John Snow Labs

04/2025 – 03/2026

(Remote) Berlin, Germany

- Lead a team of software/ML engineers, increasing release cadence to two weeks and project planning for the open source component. Maintain high code quality and reproducibility standards through code reviews, CI/CD pipelines, and technical mentoring.
- Research and integrate state-of-the-art NLP features (models, frameworks, libraries, among others) into the Spark ecosystem, coordinating with requirements from the JSL healthcare team, ensuring HIPAA compliance.
- Implement and optimize classic NLP (NER-based deidentification) and LLMs for production-ready inference for healthcare customers, achieving up to 10 times inference speedup across components.

Machine Learning and Software Engineer

John Snow Labs

04/2021 – 04/2025

(Part-time Remote) Berlin, Germany

- Integrated a custom llama.cpp engine for native, distributed on-device inference of quantized language and vision models, achieving up to 5x acceleration compared to ONNX/TensorFlow.
- Developed and optimized implementations of advanced language models (RoBERTa, UAE) and multi-modal models (CLIP, Whisper, Vision Language Models) within Spark, including critical pre-processing steps like BPE tokenization and mel-filter banks for speech-to-text models. Distributed inference achieves a linear reduction in processing time of 40% per additional node.
- Optimized the performance of existing model implementations in Spark NLP, resulting in a reduction of 50% in memory usage and 30% in processing time.
- Enabled native cross-platform inference on Apple Silicon and ARM64 architectures, expanding deployment capabilities for diverse on-device environments.

Graduate Software Engineer

Amadeus IT Group

11/2019 – 03/2021

London, United Kingdom

- Contributed to C++ based high performance back end for processing thousands of agency queries and API calls to baggage pricing providers.

Student Assistant

Distributed Artificial Intelligence (DAI) Laboratory at TU Berlin

07/2017 – 09/2019

Berlin, Germany

- Full-stack NLP development, implementing a chatbot to handle support questions for thousands of products of the Deutsche Telekom.

Education

Technische Universität Berlin

M.Sc. Computer Science, specialized in Machine Learning

2025

Berlin, Germany

Technische Universität Berlin

B.Sc. Computer Science

2019

Berlin, Germany

Skills

Languages: Python, Scala, C++, Java, Shell Script, $\text{\LaTeX}$, Markdown

Frameworks: PyTorch+PyTorch Lightning, Apache Spark, Spark NLP, llama.cpp, ONNX, TensorFlow, Tensorboard, MLLib, transformers, pandas, numpy, matplotlib

Development Tools: Git, Linux, Docker, Jupyter Notebook

Cloud Platforms: AWS, Databricks, Google Cloud Platform

Specialization: Deep Learning, Natural Language Processing, Computer Vision, Edge Inference

Spoken Languages: English (Fluent), German (Native), Vietnamese (Intermediate)

Publications

Diffusion Counterfactuals for Image Regressors: Proposed two methods to generate visual explanations for image regression models using diffusion methods. The approaches reveal which image features influence model predictions and help identify hidden biases. This paper is a condensed version of my master's thesis and was published at the peer-reviewed conference [xAI 2025](#).

Projects

Predicting Chemical Quantum Interactions

- Recreated and improved a deep learning model for [chemical quantum interactions](#).
- Developed the model from scratch in PyTorch for chemical quantum interactions, resolving critical numerical issues.
- Improved validation loss by 28% through the application of physics-informed explainability methods and contemporary regularization techniques (RMSNORM).